

LOAN APPROVAL FINANCIAL RISK ANALYSIS

1. Executive Summary

This project analyzes 20,000 loan applications to identify the key factors influencing loan approval decisions.

The analysis combines exploratory data analysis (EDA), financial risk assessment, correlation analysis, and Logistic Regression modeling to understand borrower characteristics associated with loan approval outcomes.

The results indicate that repayment capacity, income level, net worth, and credit history are among the most important factors associated with approval decisions. Conversely, high debt burdens, large loan amounts, and elevated interest rates significantly reduce approval likelihood.

2. Business Problem

Financial institutions must evaluate loan applications while balancing profitability and credit risk.

The objective of this project is to identify which client profiles are more likely to receive loan approval and determine the key financial and risk-related factors associated with approval decisions.

Key business questions include:

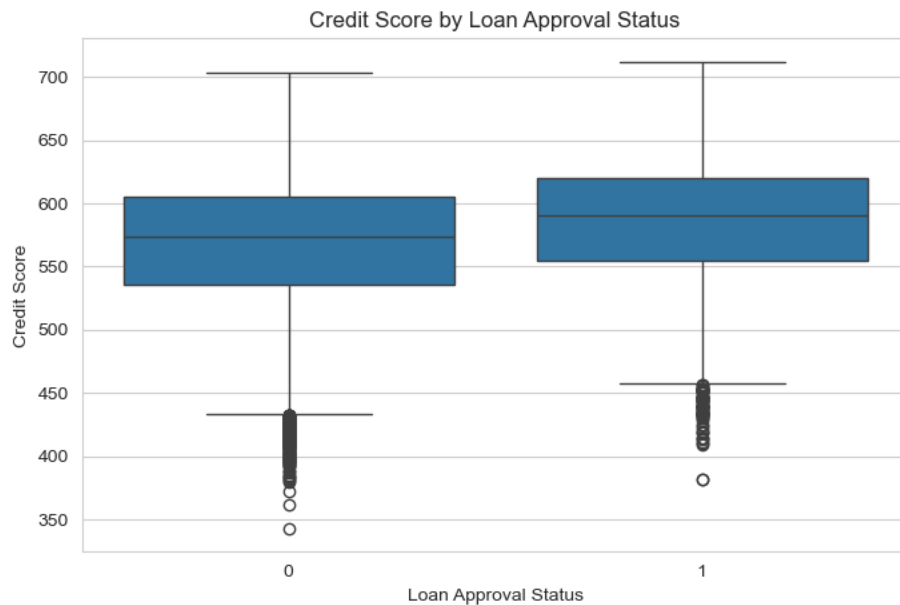
- What characteristics distinguish approved applicants from rejected applicants?
- Which financial indicators are associated with loan approval?
- How does borrower risk affect approval outcomes?
- Which factors have the strongest influence on approval decisions?

3. Dataset Overview

Metric	Value
Records	20,000
Features	36
Missing Values	0
Duplicates	0

4. Key Insights

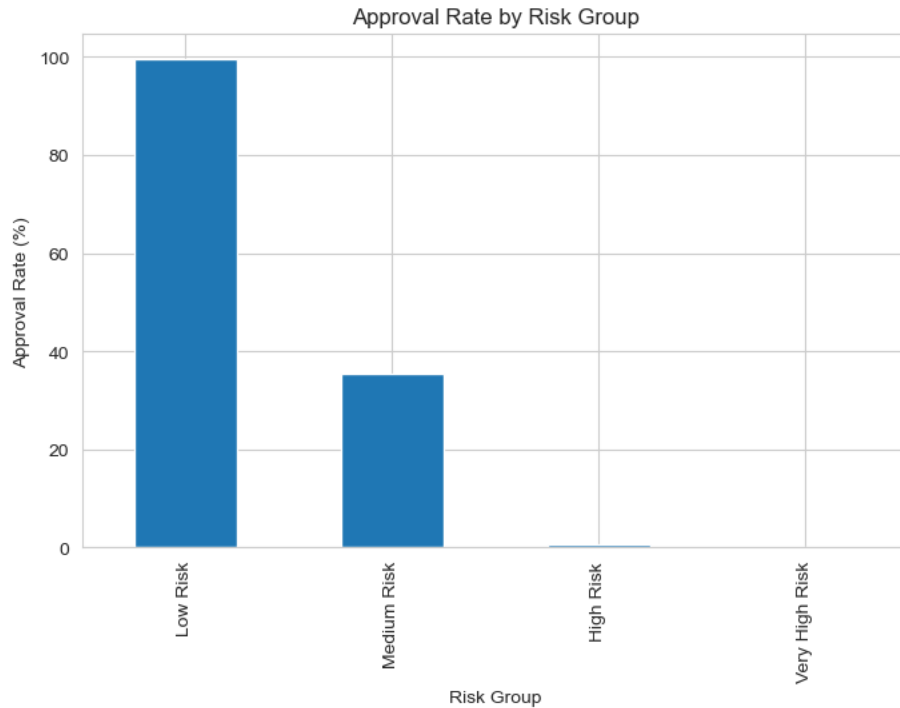
Insight 1: Credit Score



Approved applicants generally have higher credit scores than rejected applicants. The median credit score of approved borrowers is noticeably higher, suggesting that creditworthiness contributes positively to loan approval decisions.

However, the overlap between the two distributions indicates that credit score alone is not sufficient to explain approval outcomes.

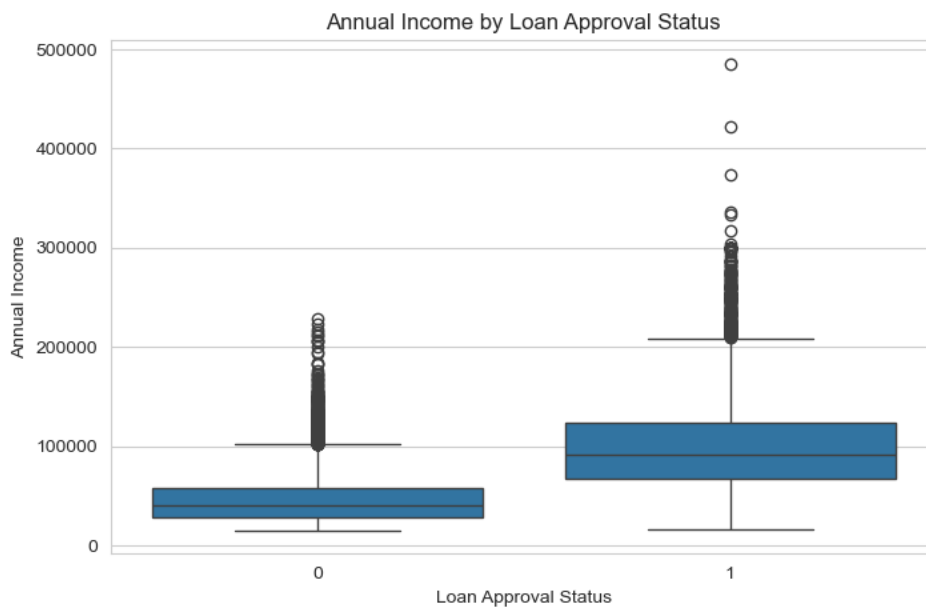
Insight 2: Risk Group



Loan approval rates decline sharply as risk scores increase.

Nearly all low-risk applicants receive approval, while approval rates drop dramatically among high-risk and very high-risk borrowers. This suggests that risk assessment is one of the most important factors considered during the approval process.

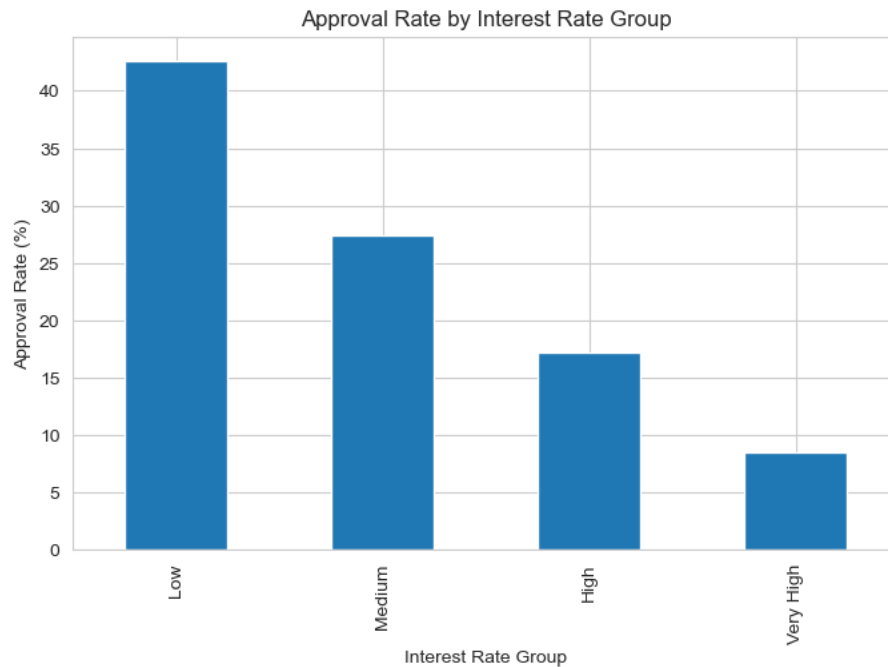
Insight 3: Income



Approved applicants earn substantially higher incomes than rejected applicants.

This finding suggests that income level plays a significant role in loan approval decisions, as higher-income borrowers are generally perceived to have stronger repayment capacity and lower default risk.

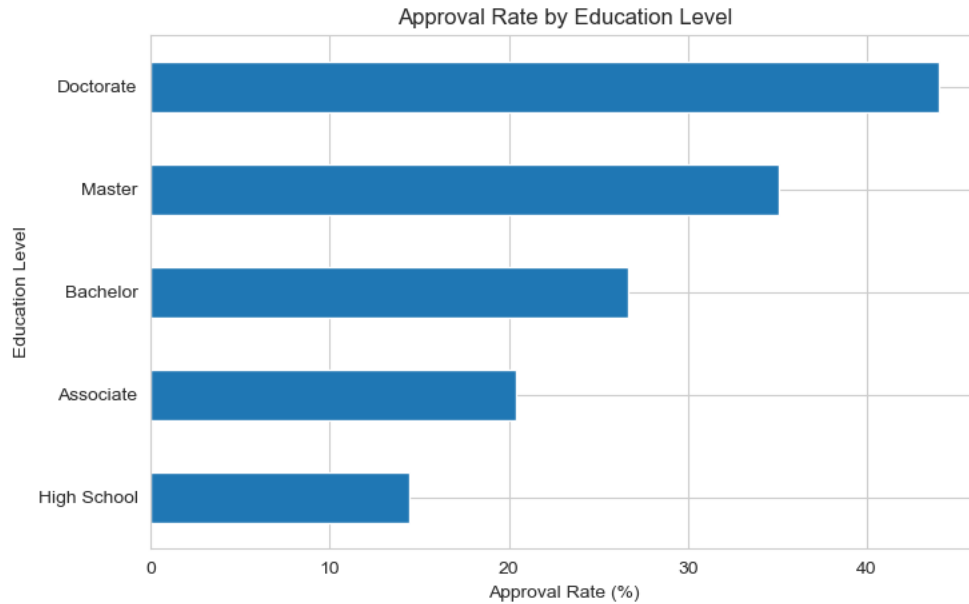
Insight 4: Interest Rate



Applicants associated with lower interest rate groups exhibit considerably higher approval rates.

Approval probability decreases as interest rates increase, suggesting that borrowers offered lower-priced loans typically possess stronger financial profiles and lower perceived credit risk.

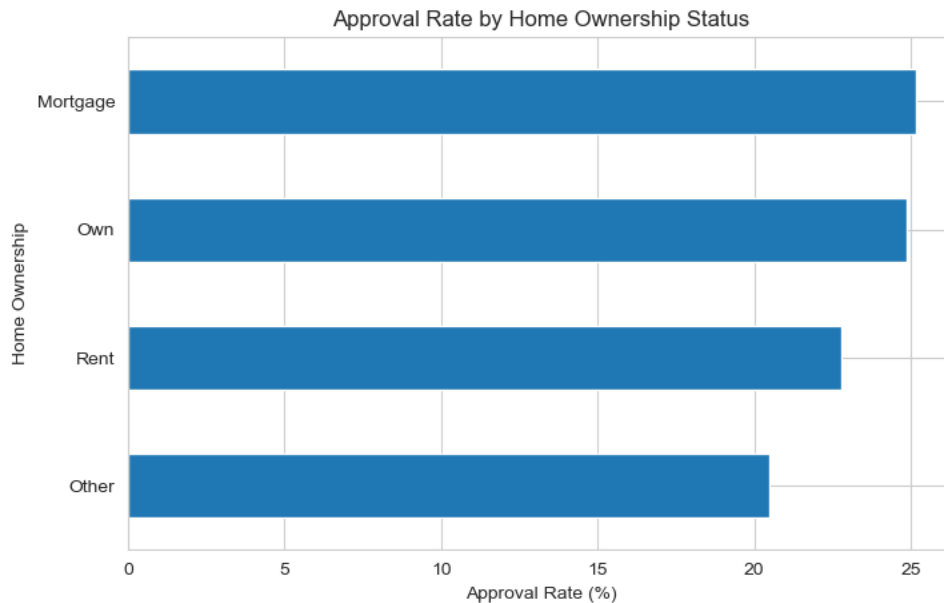
Insight 5: Education Level



Loan approval rates increase consistently with educational attainment.

Applicants holding Master's and Doctoral degrees achieve noticeably higher approval rates compared with those holding only a High School diploma. This pattern suggests that higher education levels may be associated with stronger financial stability and income potential.

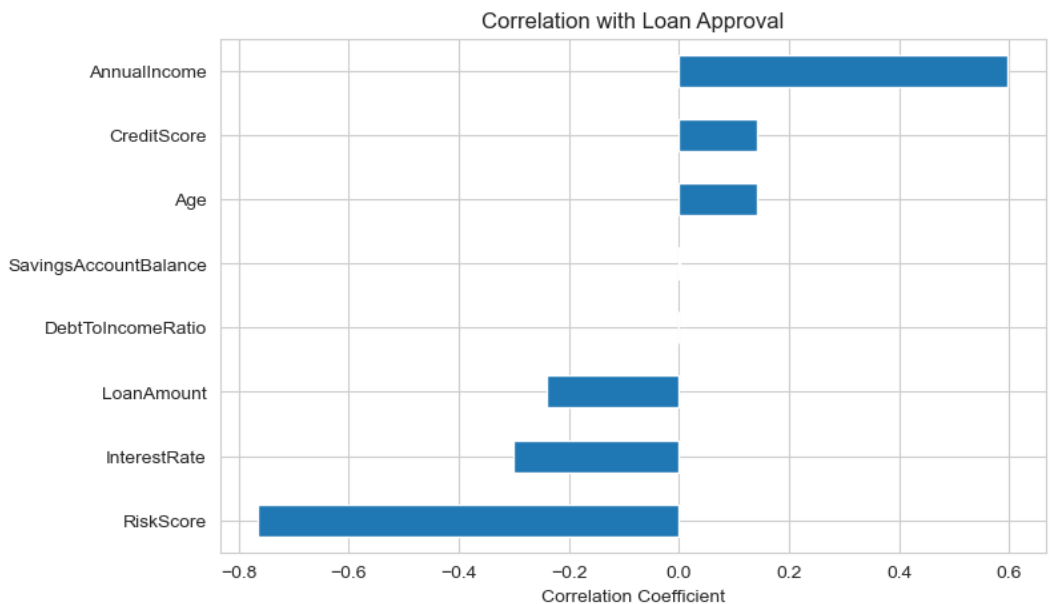
Insight 6: Home Ownership



Approval rates vary only slightly across home ownership categories.

Although applicants with mortgages and owned properties demonstrate marginally higher approval rates, housing status appears to have a relatively limited influence on loan approval outcomes compared with other financial variables.

5. Correlation Analysis



The correlation analysis reveals that RiskScore exhibits the strongest relationship with loan approval, followed by AnnualIncome.

InterestRate and LoanAmount are negatively associated with approval outcomes, indicating that larger and more expensive loans are less likely to be approved. Meanwhile, SavingsAccountBalance and Debt-to-Income Ratio show minimal correlation with approval decisions within this dataset.

6. Logistic Regression

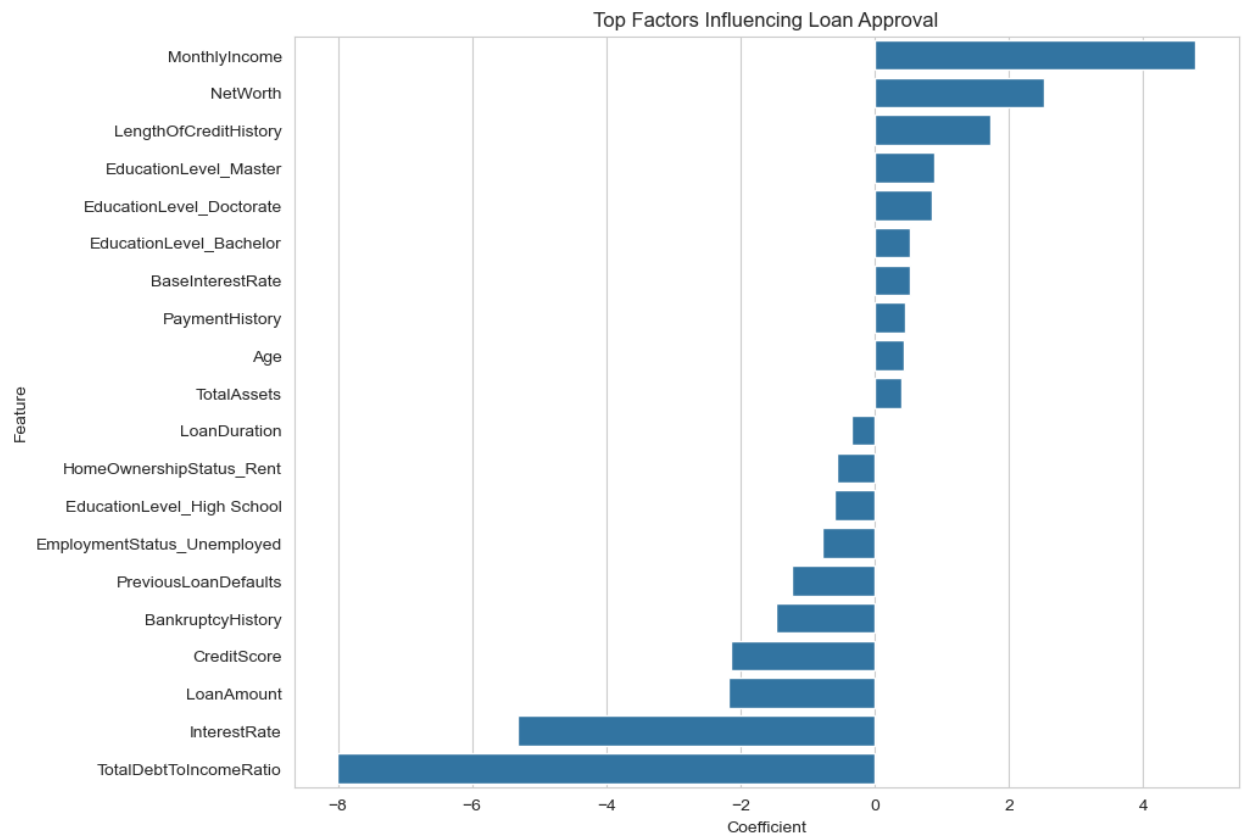
Model Performance

Metric	Value
Accuracy	96.75%
Precision	94%
Recall	93%
F1-score	93%

A Logistic Regression model was developed to identify the factors most strongly associated with loan approval outcomes.

To reduce target leakage, RiskScore was excluded from the final model. The model achieved strong predictive performance, demonstrating that borrower financial characteristics provide substantial explanatory power for approval decisions.

7. Key Drivers



The Logistic Regression model indicates that Monthly Income, Net Worth, and Length of Credit History are the strongest positive drivers of loan approval.

Conversely, higher Debt-to-Income Ratio, Interest Rate, and Loan Amount are associated with lower approval probabilities. These findings suggest that applicants with stronger financial capacity and longer credit histories are more likely to receive loan approval.

8. Business Recommendations

Recommendation 1

Focus marketing efforts on applicants with higher income levels and stronger financial profiles, as they demonstrate significantly higher approval rates.

Recommendation 2

Strengthen risk assessment procedures for applicants with high Risk Scores and large debt burdens to reduce potential default risk.

Recommendation 3

Review loan approval criteria for applicants with strong credit histories but lower approval rates to identify potential growth opportunities.

9. Conclusion

This project examined 20,000 loan applications to identify the key factors associated with loan approval decisions.

The analysis revealed that borrower financial strength plays a significant role in approval outcomes. Applicants with higher income levels, greater net worth, and longer credit histories were more likely to receive approval, while larger loan amounts, higher interest rates, and heavier debt burdens reduced approval probability.

The Logistic Regression model achieved strong predictive performance, demonstrating that borrower financial characteristics provide substantial explanatory power for loan approval decisions. Overall, the findings highlight the importance of affordability, financial stability, and creditworthiness in lending decisions.